

Geopolitical Sentiment & Semiconductor Equity Risk

A Cross-Platform, Multi-Model NLP Pipeline: Reddit & Bluesky → NVDA · AMD · INTC

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MOTIVATION & RESEARCH QUESTIONS

U.S. export-control policy—Bureau of Industry and Security (BIS) rules, Entity List additions, and technology sanctions targeting China—creates measurable uncertainty for semiconductor firms. Online uncertainty in real time, ahead of market-moving events.

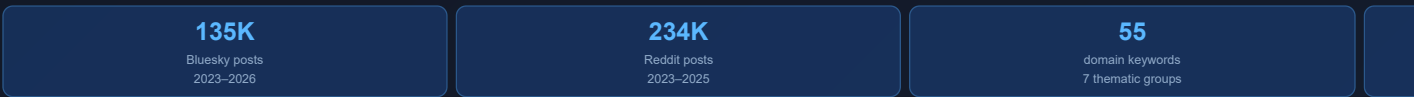
Core Question

Does geopolitical / export-control sentiment extracted from social media contain incremental explanatory power for semiconductor stock returns and volatility beyond standard market factors?

Three Formal Hypotheses

- **H1 (Risk Premium):** Negative policy-risk sentiment predicts lower abnormal returns around policy announcement windows.
- **H2 (Volatility):** Elevated policy sentiment today forecasts higher realized volatility over the next 1–5 trading days.
- **H3 (Exposure Heterogeneity):** Effects are stronger for high-exposure firms (NVDA, ASML) than for lower-exposure peers.

DATA & PIPELINE



Three Sentiment Models (applied identically on both platforms)

- **NRC Lexicon** — Emotion dimensions: joy, fear, anger, trust, anticipation, sadness, disgust, surprise + net valence.
- **HuggingFace (EN)** — 4-class transformer: very_pos, pos, neg, very_neg → net score.
- **HuggingFace (Multilingual)** — Same architecture; handles non-English posts (~5.6% of Reddit, ~0% Bluesky after filter).
- **FinBERT** — Finance-domain fine-tuned BERT: positive / neutral / negative probabilities.

Policy-Context Tiering

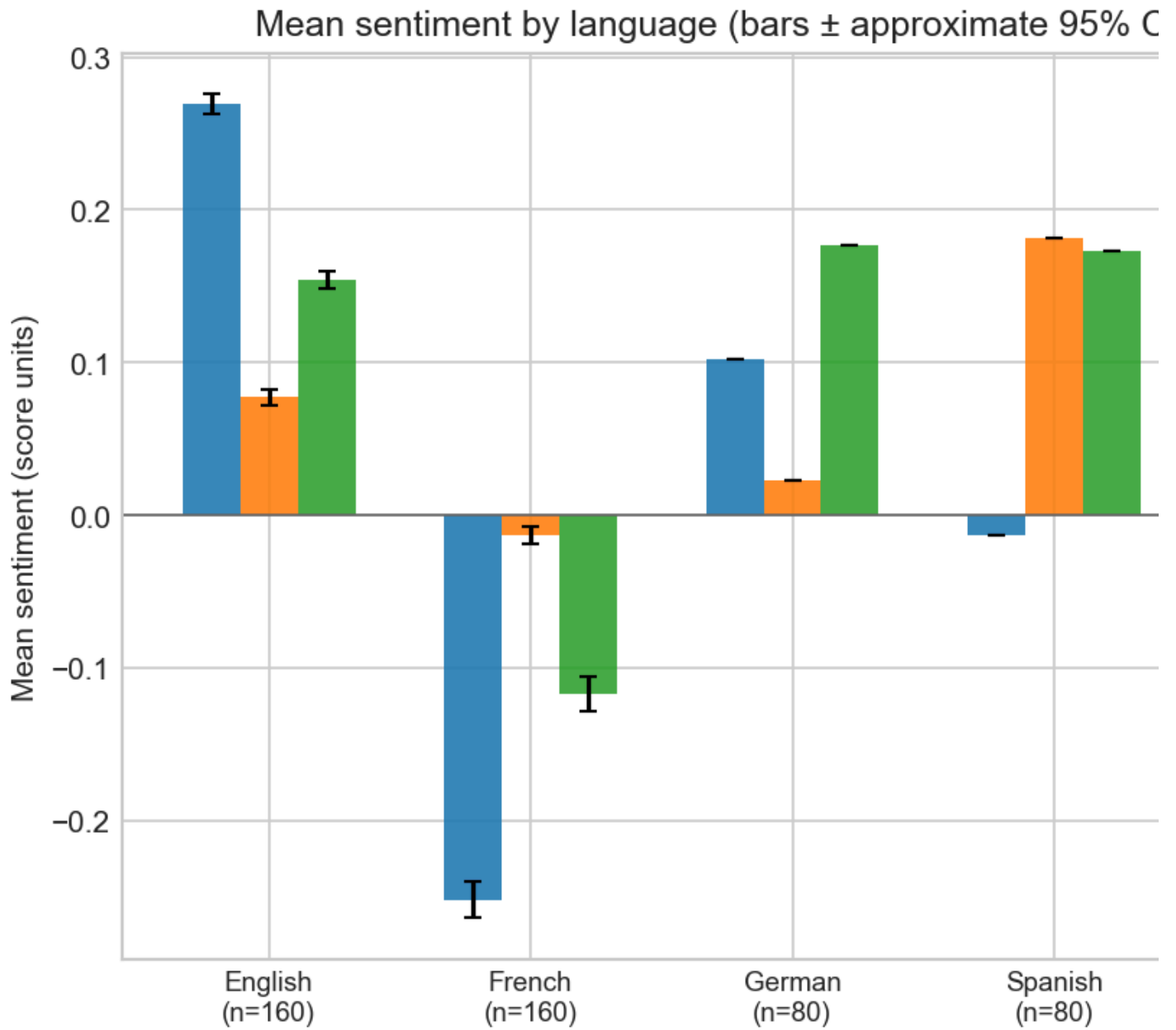
Tier 1 (policy-relevant, 14.6% of Bluesky): posts containing BIS, Entity List, export ban, China, Taiwan, sanction, Huawei, SMIC, etc. **Tier 2** (general firm sentiment). Both tiers are analyzed separately.

Look-Ahead Bias Control

All posts published after 4:00 PM ET are assigned to the *next* NYSE trading day. Sentiment on day *t* predicts returns only from *t*+1 onward.

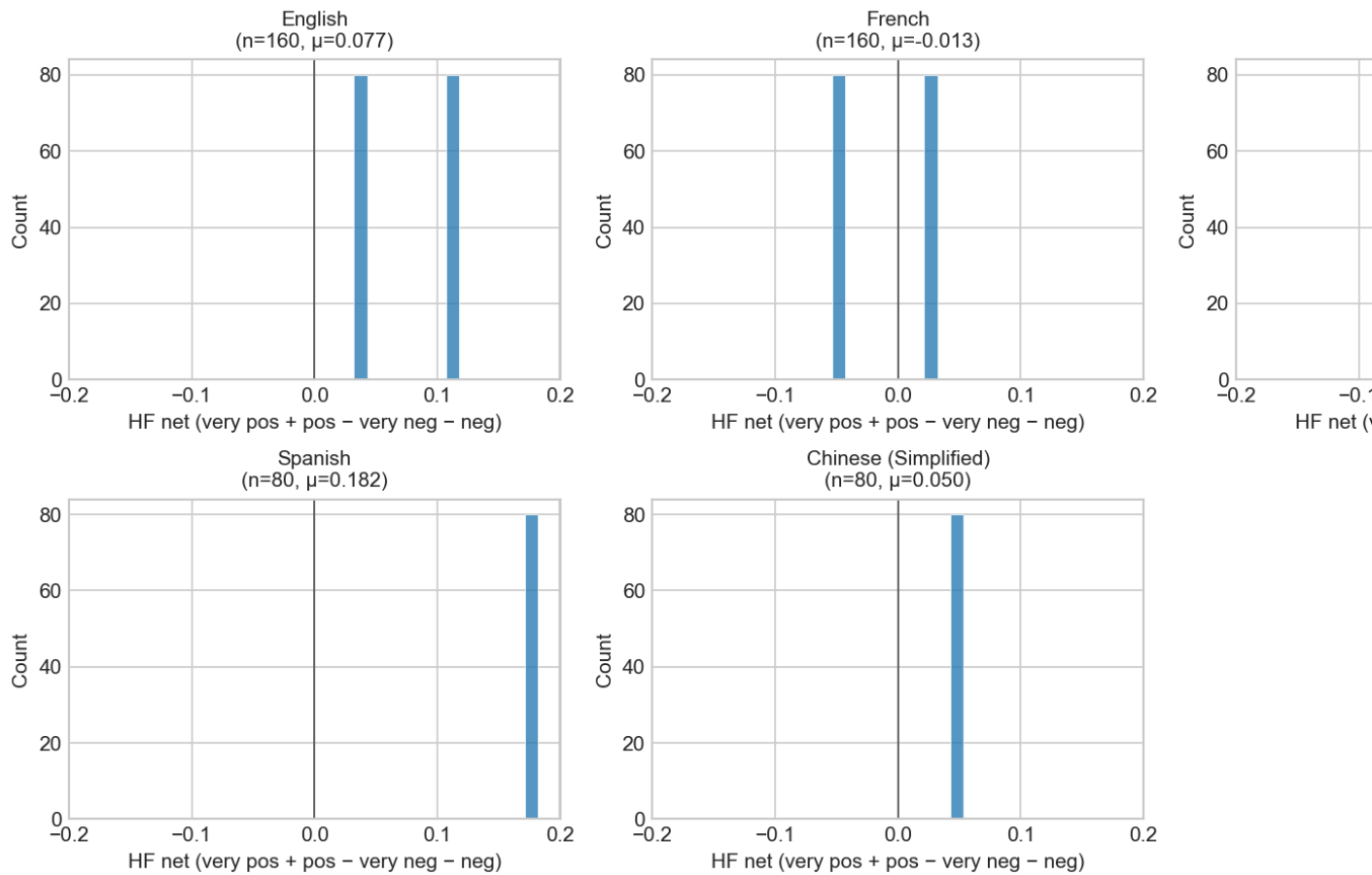
CROSS-LANGUAGE SENTIMENT (REDDIT)

Reddit contains ~5.6% non-English posts (de, fr, nl, it, pt, sv, ca). HuggingFace multilingual model scores these natively without translation.



Mean HF sentiment by detected language — Reddit

Distribution of HuggingFace multilingual net sentiment by language



HF net score distribution stratified by language

Non-English posts show similar sentiment distributions to English posts, suggesting cross-language robustness of the multilingual scoring pipeline.